

AI System Specification

Course: Introduction to Artificial Intelligence

Topic: PEAS Framework and Task Environment Analysis

Reference: Russell & Norvig — Artificial Intelligence: A Modern Approach, Chapter 2

Student Name	Ghulam Kabir	Roll Number	2K23/CSM/40
Course	Introduction to Artificial Intelligence	Option	D — Medical Imaging Assistant

Task 1 · PEAS Specification

The **Medical Imaging Assistant** is an AI-powered diagnostic support system deployed in hospital radiology departments. It analyzes radiological images — including X-rays, MRI scans, and CT scans — to automatically detect abnormalities, generate structured diagnostic reports, and prioritize critical cases for physician review.

Performance Measures	1. Diagnostic Accuracy — Overall percentage of pathologies correctly identified; clinically acceptable threshold is greater than 95% 2. False Negative Rate (FNR) — Percentage of actual diseases the system fails to detect; most critical metric since a missed disease leads to delayed treatment 3. False Positive Rate (FPR) — Percentage of healthy cases incorrectly flagged; high FPR causes unnecessary procedures and patient anxiety 4. Processing Speed — Time to fully analyze one radiological scan; target is under 30 seconds per image for real-time clinical workflows 5. Sensitivity — Proportion of actual positive cases correctly identified (True Positive Rate) 6. Specificity — Proportion of actual negative cases correctly cleared (True Negative Rate)
Environment	Hospital radiology department with DICOM-based medical imaging infrastructure including PACS servers, EHR systems, and radiology workstations. Key External Factors: - Image quality variation across different scanner manufacturers and models - Diverse patient demographics including age, gender, and body composition - Variable imaging protocols, slice thicknesses, and contrast agent dosages - High-pressure emergency clinical environments with strict time constraints - Healthcare data privacy regulations (HIPAA and local laws)

Actuators	1. Visual Overlay Display — Renders color-coded bounding boxes and heatmaps on radiological images to highlight detected abnormal regions for radiologist review 2. Automated Report Generator — Produces a structured PDF diagnostic report with findings, confidence scores, and recommended follow-up actions for the physician 3. Priority Alert System — Flags and escalates critical cases such as suspected hemorrhage or large tumors to the top of the radiology review queue 4. EHR Integration Module — Writes diagnostic findings and AI confidence scores directly into the patient Electronic Health Record for longitudinal tracking 5. Second Opinion Trigger — Automatically requests human radiologist review when the system's confidence score falls below a defined safety threshold
Sensors	1. High-Resolution X-ray Detector — Captures 2D grayscale radiographs of skeletal structures and chest cavity at resolutions up to 4096 x 4096 pixels 2. MRI Scan Slice Input (DICOM) — Receives T1/T2-weighted multi-plane anatomical image slices for detailed soft tissue analysis 3. CT Scan 3D Volume Data — Receives multi-slice volumetric cross-sectional data for tumor localization and organ analysis 4. Patient Metadata Input — Structured EHR data including age, gender, symptoms, prior diagnoses, and current medications 5. Contrast Agent Sensor — Detects and processes contrast-enhanced regions for vascular and cardiac imaging studies 6. Radiologist Feedback Input — Captures corrections and annotations from radiologists to enable continuous model improvement through active learning

Task 2 · Environment Classification

The task environment of the Medical Imaging Assistant is classified across the six standard dimensions defined by Russell & Norvig. Each classification includes a detailed justification specific to this system.

Dimension	Classification	Justification
Fully Observable vs. Partially Observable	PARTIALLY OBSERVABLE	The system receives only radiological images and limited patient metadata available in the EHR at the time of analysis. Complete clinical history, current lab results, physical examination findings, and verbal symptom descriptions are not accessible. This incomplete view requires the system to make probabilistic inferences based on partial information.
Single-Agent vs. Multi-Agent	SINGLE- AGENT	The Medical Imaging Assistant operates as one autonomous AI agent that independently processes each imaging case. Human radiologists retain final clinical decision-making authority but are considered external authorities, not co-agents within the AI system. There are no competing or cooperating AI agents involved in the same task.
Deterministic vs. Stochastic	STOCHASTIC	The same underlying pathology can produce visually different images depending on scanner calibration, patient positioning and motion artifacts, imaging protocol variations, and patient-specific biological differences. Because the system cannot fully predict its output from a given input due to these uncontrollable variables, the environment is stochastic. Probabilistic models and confidence scores are therefore essential.
Episodic vs. Sequential	SEQUENTIAL	Accurate diagnosis frequently requires comparison with the patient's prior scans to identify disease progression or regression over time. For example, detecting whether a lung nodule has grown requires comparing current and historical CT scans. Current output directly influences how future scans will be interpreted for the same patient.
Static vs. Dynamic	STATIC	Once a radiological image is captured and submitted for analysis, it does not change during the time the AI is processing it. The image remains a fixed, static input throughout the entire inference pipeline from preprocessing to diagnosis output. While the broader clinical context may evolve, the specific image under analysis at any given moment is stable.
Discrete vs. Continuous	CONTINUOUS	Both inputs and outputs of this system are continuous. Input images consist of pixel intensity values across a continuous spectrum (0–65535 in 16-bit imaging). Output confidence scores, disease probability estimates, bounding box coordinates, and segmentation masks are all expressed as real-valued continuous numbers.

Task 3 · Critical Analysis

Question 1 — Which dimension poses the greatest technical challenge and why?

Among the six environment dimensions, the **Stochastic** nature of the Medical Imaging Assistant's environment presents the greatest and most complex technical challenge for an AI engineer.

In deterministic systems, the same input always produces the same output, making model training and validation straightforward. However, in medical imaging this is never the case. Consider the following real-world sources of stochasticity:

Scanner Variability	A Siemens MRI machine and a GE MRI machine may produce structurally different images of the same patient's brain even at identical field strength and protocol settings.
Motion Artifacts	Involuntary movement during scanning such as breathing and heartbeat introduces blurring and noise that varies unpredictably between patients.
Biological Variation	The same disease at the same stage can appear dramatically different in a 25-year-old versus a 70-year-old due to natural anatomical and tissue differences.
Protocol Differences	Different hospitals use different slice thicknesses, contrast dosages, and imaging sequences, all of which alter the visual appearance of pathologies.

These factors create the **domain shift problem** — a model trained at one hospital may perform poorly when deployed at another with different equipment. To address this, engineers must collect large diverse multi-institutional training datasets, apply data augmentation techniques, use domain adaptation and transfer learning methods, and conduct rigorous external validation studies before any clinical deployment.

Question 2 — Propose a Utility Function to handle a trade-off

In clinical environments, the Medical Imaging Assistant must balance two competing objectives: maximizing **Sensitivity** (catching all real disease cases) while minimizing the **False Positive Rate** (avoiding unnecessary false alarms). Missing a real disease is far more dangerous than a false alarm in medical settings.

Primary Utility Function — Sensitivity vs. False Positive Rate:

$$U = 0.7 \times \text{Sensitivity} - 0.3 \times \text{False_Positive_Rate}$$

Weight: 0.7	Sensitivity is weighted higher because a missed diagnosis (false negative) can result in delayed treatment and serious patient harm. A patient whose cancer goes undetected may not receive timely treatment, potentially resulting in irreversible harm or death.
Weight: 0.3	False Positive Rate is penalized to prevent the system from becoming overly aggressive and generating excessive unnecessary alerts that overwhelm clinical staff and lead to unnecessary biopsies or procedures.

Effect of Doubling the Sensitivity Weight:

If the Sensitivity weight were doubled from **0.7 to 1.4**, the agent's behavior would shift dramatically:

Aggressive Flagging	The agent would flag any region with even a small probability of being abnormal, reducing missed diagnoses to near zero.
Increased False Alarms	The False Positive Rate would increase significantly, generating many more incorrect alerts per scan.
Higher Workload	Radiologists would face a much heavier review burden, potentially slowing down the overall clinical workflow.
Best For	Mass cancer screening programs such as national lung cancer screening where catching every possible case is the top priority even at the cost of additional follow-up procedures.

Task 4 · Structured Data Design — JSON Practice

The following valid JSON object encodes the complete PEAS specification and environment classification for the Medical Imaging Assistant as a single machine-readable structured format. This representation enables integration into software pipelines, documentation systems, and AI specification databases.

Structured Representation

```
{
  "system_name"      : "Medical Imaging Assistant for Early Disease Detection",
  "student"          : "Ghulam Kabir",
  "roll_number"      : "2K23/CSM/40",
  "course"           : "Introduction to Artificial Intelligence",
  "application_option": "D",

  "peas": {

    "performance_measures": [
      { "metric": "diagnostic_accuracy", "target": "> 95%"      },
      { "metric": "false_negative_rate", "target": "minimize"  },
      { "metric": "false_positive_rate", "target": "minimize"  },
      { "metric": "scan_processing_speed", "target": "< 30 seconds" },
      { "metric": "sensitivity",          "target": "maximize"  },
      { "metric": "specificity",          "target": "maximize"  }
    ],

    "environment": {
      "location"          : "Hospital radiology department",
      "infrastructure"     : "DICOM-based imaging systems – PACS, EHR, radiology work stations",
      "imaging_modalities": ["X-ray", "MRI", "CT Scan"],
      "external_factors"  : [
        "image quality variation across scanner manufacturers and models",
        "diverse patient demographics and body compositions",
        "variable imaging protocols and contrast agent dosages",
        "high-pressure emergency clinical environments",
        "healthcare data privacy regulations (HIPAA)"
      ]
    }
  },

  "actuators": [
    "visual_overlay_display_bounding_boxes_and_heatmaps",
    "automated_structured_diagnostic_pdf_report_generator",
    "priority_alert_escalation_system_for_critical_cases",
    "ehr_integration_module_write_findings",
    "second_opinion_request_trigger_low_confidence_cases"
  ],

  "sensors": [
    "high_resolution_digital_xray_detector_4096x4096",
    "mri_scan_slice_input_t1_t2_weighted_dicom_format",
    "ct_scan_3d_volumetric_data_multi_slice",
    "patient_metadata_age_gender_symptoms_history_ehr",
    "contrast_agent_detection_sensor_vascular_imaging",
  ]
}
```

```

    "radiologist_feedback_and_annotation_input"
  ]
},

"environment_classification": {
  "observability" : {
    "choice"      : "Partially Observable",
    "justification" : "Only images and limited EHR metadata available; full clinical history absent"
  },
  "agents" : {
    "choice"      : "Single-Agent",
    "justification" : "One AI model per case; human radiologist is external authority not a co-agent"
  },
  "determinism" : {
    "choice"      : "Stochastic",
    "justification" : "Same pathology appears differently due to scanner variation and motion artifacts"
  },
  "episodicity" : {
    "choice"      : "Sequential",
    "justification" : "Diagnosis depends on prior scans and longitudinal patient history"
  },
  "dynamism" : {
    "choice"      : "Static",
    "justification" : "Radiological image is fixed during inference; environment does not change"
  },
  "continuity" : {
    "choice"      : "Continuous",
    "justification" : "Pixel values and confidence scores are real-valued continuous data"
  }
},

"utility_function": {
  "formula" : "U = 0.7 * Sensitivity - 0.3 * False_Positive_Rate",
  "weights" : { "sensitivity": 0.7, "false_positive_rate": -0.3 },
  "purpose" : "Balance disease detection sensitivity with false alarm reduction"
}
}

```